

Questions with Answer Keys

MathonGo

Q1 - 2024 (27 Jan Shift 1)

If the shortest distance of the parabola $y^2 = 4x$ from the centre of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$ is d , then d^2 is equal to :

- (1) 16
- (2) 24
- (3) 20
- (4) 36

Q2 - 2024 (29 Jan Shift 2)

Let $P(\alpha, \beta)$ be a point on the parabola $y^2 = 4x$. If P also lies on the chord of the parabola $x^2 = 8y$ whose mid point is $(1, \frac{5}{4})$. Then $(\alpha - 28)(\beta - 8)$ is equal to _____.

Q3 - 2024 (30 Jan Shift 1)

The maximum area of a triangle whose one vertex is at $(0, 0)$ and the other two vertices lie on the curve $y = -2x^2 + 54$ at points (x, y) and $(-x, y)$ where $y > 0$ is :

- (1) 88
- (2) 122
- (3) 92
- (4) 108

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Questions with Answer Keys

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Solutions

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Q1

Equation of normal to parabola

$$y = mx - 2m - m^3$$

this normal passing through center of circle (2, 8)

$$8 = 2m - 2m - m^3$$

$$m = -2$$

So point P on parabola $\Rightarrow (am^2, -2am) = (4, 4)$

And C = (2, 8)

$$PC = \sqrt{4 + 16} = \sqrt{20}$$

$$d^2 = 20$$

Q2

Parabola is $x^2 = 8y$ Chord with mid point (x_1, y_1) is T = S₁

$$\therefore xx_1 - 4(y + y_1) = x_1^2 - 8y_1$$

$$\therefore (x_1, y_1) = \left(1, \frac{5}{4}\right)$$

$$\Rightarrow x - 4\left(y + \frac{5}{4}\right) = 1 - 8 \times \frac{5}{4} = -9$$

$$\therefore x - 4y + 4 = 0 \dots (i)$$

 (α, β) lies on (i) & also on $y^2 = 4x$

$$\therefore \alpha - 4\beta + 4 = 0 \dots (ii)$$

$$\&\beta^2 = 4\alpha \dots (iii)$$

Solving (ii) & (iii)

$$\beta^2 = 4(4\beta - 4) \Rightarrow \beta^2 - 16\beta + 16 = 0$$

$$\therefore \beta = 8 \pm 4\sqrt{3} \text{ and } \alpha = 4\beta - 4 = 28 \pm 16\sqrt{3}$$

$$\therefore (\alpha, \beta) = (28 + 16\sqrt{3}, 8 + 4\sqrt{3}) \text{ \& } (28 - 16\sqrt{3}, 8 - 4\sqrt{3})$$

$$(28 - 16\sqrt{3}, 8 - 4\sqrt{3})$$

$$\therefore (\alpha - 28)(\beta - 8) = (\pm 16\sqrt{3})(\pm 4\sqrt{3})$$

$$= 192$$

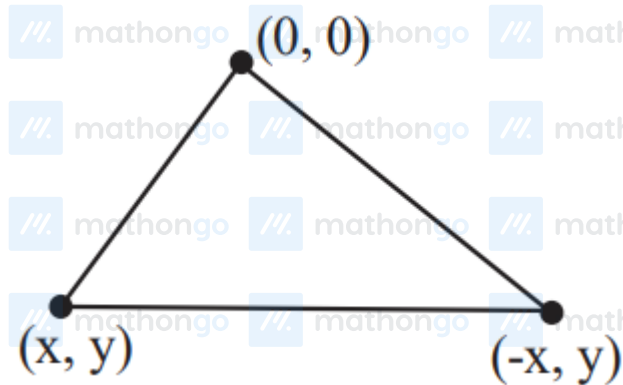
Q3

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Solutions

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Area of Δ

$$= \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ x & y & 1 \\ -x & y & 1 \end{vmatrix}$$

$$\Rightarrow \left| \frac{1}{2}(xy + xy) \right| = |xy|$$

$$\text{Area}(\Delta) = |xy| = |x(-2x^2 + 54)|$$

$$\frac{d(\Delta)}{dx} = |(-6x^2 + 54)| \Rightarrow \frac{d\Delta}{dx} = 0 \text{ at } x = 3$$

$$\text{Area} = 3(-2 \times 9 + 54) = 108$$

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